Lecture 5-2: OLS Assumptions

Goal: Examine OLS assumptions, omitted variable bias, and multicollinearity.

- Multiple OLS assumptions
  - Linearity in parameters
  - Random sampling
  - No perfect collinearity
  - Zero conditional mean
  - Homoskedasticity
  - Normality
- Omitted variable bias
  - Excluding a relevant variable
  - Bias direction depends on correlations
  - Confounders defined
- Multicollinearity
  - Variance of OLS estimators & estimation
  - Detecting multicollinearity
  - Variance inflation factor
- Including irrelevant variables
  - No bias, but larger variance